(1.) LOOKUP Function :- The LOOKUP function searches for a value in a single row or column and returns a corresponding value from another row or column.

(*) Syntax :- =LOOKUP (lookup_value, lookup_vector, result_vector).

(*) Example :

Employee ID	Employee Name
101	Rahul
102	Priya
103	Divya
104	Kiran

=LOOKUP (103, A2:A5, B2:B5). (To Find the employee name for ID 103).
o/p : Divya.

(*) Advantages of LOOKUP :-
- Simple to use for basic lookups.
- can search horizontally or vertically.
- Useful for retrieving corresponding values from a range.

(*) Disadvantages of LOOKUP :-
- Works best when lookup values are sorted in ascending order.
- May return an approximate match rather than an exact match.
- Less flexible than modern lookup functions.

(2.) VLOOKUP Function :- VLOOKUP stands for vertical Lookup. It searches for a value in the first column of a table & returns a related value from another column in the same ~~to~~ row.

(*) Syntax : =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup]).

(*) Example :-

Employee ID	Employee Name	Salary.
101	Rahul	25000
102	Priya	30000
103	Divya	35000
104	Kiran	28000

To find priya's salary using Employee ID 102.
=VLOOKUP(102, A2:C5, 3, 0).
o/p: 30000.

0 → Searches for exact match

1 means for nearest match

(*) Advantages of VLOOKUP Function :-
- Easy to learn & widely used.
- Retrieves information from large tables.
- Supports exact & approximate matching.

(*) Disadvantages of VLOOKUP Function :-
- Searches only in the first column of the selected table.
- Cannot directly return a value from a column to the left of the lookup column.
- Column numbers may need updating if the table structure changes.

(3.) HLOOKUP Function:- HLOOKUP stands for Horizontal LOOKup. It searches for a value in the first row of a table & returns a related value from a specified row in the same column.

(*) Syntax:- =HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup]).

(*) Example: Suppose your employee data is arranged horizontally:

	Column B	Column C	Column D
Employee ID	101	102	103
Employee Name	Rahul	Priya	Divya
Salary	25000	30000	35000

To find the salary of employee ID 102:
=HLOOKUP(102, B1:D3, 3, FALSE).
o/p - 30,000.

(*) Advantages of HLOOKUP:-
- Useful when data is arranged horizontally.
- Easy to retrieve information from rows.
- Supports exact and approximate matching.

(*) Disadvantages of HLOOKUP:-
- Searches only in the first row of the selected table.
- Cannot directly return a value from a row above the lookup row.
- Can become difficult to maintain when rows are inserted or rearranged.